

Comparative Hardware Architecture and System Topologies of SFF and NUC PCs for Creative Workstations

Source guide

This report explores the architectural evolution of small form factor (SFF) and NUC computers, detailing how they have transformed from simple office tools into high-performance creative workstations. It categorizes the industry by examining three primary ecosystems—Apple’s unified memory systems, AMD’s AI-focused mobile processors, and Intel’s hybrid architectures—while weighing the trade-offs between thermal efficiency and raw computational power. A central theme is the use of high-bandwidth interfaces like OCuLink and Thunderbolt to overcome physical size constraints, allowing these compact units to utilize external desktop-grade GPUs for intensive 3D and video tasks. Ultimately, the text serves as a strategic guide for professionals to configure specialized setups for 4K video editing, real-time rendering, and dual-PC live broadcasting using modern mini-PC hardware.

The landscape of professional content creation is undergoing a structural transition as small form factor (SFF) computers and Next Unit of Computing (NUC) systems emerge as genuine alternatives to traditional full-sized desktop towers. Historically, resource-intensive workflows such as real-time 4K/8K video editing, complex 3D rendering, local artificial intelligence (AI) inference, and multi-channel live broadcasting demanded expansive computer chassis to accommodate power-hungry components and massive cooling solutions.[1, 2, 3] However, microarchitectural advancements, unified memory systems, and high-bandwidth external interfaces have converged to deliver desktop-class performance within enclosures measuring under three liters in volume.[4, 5, 6, 7]

This transition is driven by a combination of thermal efficiency and modular expandability.[1, 8, 9] By offloading graphics computation to external desktop-grade hardware over direct PCIe connections, and leveraging dedicated hardware-accelerated media encoders, modern mini PCs can successfully execute workflows that were previously restricted to enterprise-grade workstation towers.[8, 10, 11] This report analyzes the primary hardware platforms, external expansion technologies, and network setups that define the state of the art for content creators utilizing mini and NUC PCs.

Architectural Foundations of SFF Platforms in Professional Workflows

To select the appropriate SFF hardware, creative professionals must evaluate several distinct platform architectures, each presenting unique trade-offs in memory bandwidth, expansion capability, and hardware-level media encoding.

Apple Silicon Ecosystem and Media Acceleration

The introduction of the Apple Mac mini equipped with the M4 and M4 Pro system-on-chip (SoC) architectures represents a major milestone in SFF computational density.[5, 12] At the core of Apple’s performance advantage is its Unified Memory Architecture (UMA).[5, 13] Unlike traditional architectures where the central processing unit (CPU) and discrete graphics processing unit (GPU) maintain separate memory pools and transfer data across a bottlenecked

PCIe bus, Apple's UMA allows all system processing engines to access a single, high-bandwidth pool of RAM.[5, 13] On the M4 Pro chip, this memory bandwidth reaches up to 273 GB/s, which dramatically accelerates asset-heavy creative workloads such as multi-cam editing and high-resolution texture mapping.[3, 11] Crucially for video editors, the M4 and M4 Pro SoCs feature a highly specialized, hardware-accelerated Media Engine.[5, 11] This silicon block contains dedicated hardware decoders and encoders for common professional video formats, including H.264, HEVC, ProRes, and ProRes RAW, alongside AV1 hardware decoding.[5, 11] The physical impact of these media engines is illustrated in real-world benchmarks where the Mac mini M4 Pro completes a Premiere Pro 4K export in 3 minutes and 20 seconds, outperforming more expensive mobile systems and rivaling enterprise workstations in specific rendering tasks.[3]

System Configuration	Resolve Timeline Playback (Neat Video Pro 6)	Resolve Timeline Playback (Native UltraNR)	Multi-cam Stream Limit (Before Frame Drops)	Render Speed Improvement (vs. Prior Gen)
Apple Mac mini M4 Pro	Realtime (30 FPS) [14]	Limited (17 FPS) [14]	10 to 15 Streams [3]	+49% [3]
M1 Max MacBook Pro 16	~14.28 FPS [14]	N/A	N/A	Baseline
M1 Ultra Mac Studio	~23.14 FPS [14]	N/A	N/A	N/A

Despite these capabilities, the architecture enforces a rigid constraint: the unified memory and solid-state storage are soldered directly to the package.[5, 12, 15] Because post-purchase memory upgrades are impossible, creators must carefully specify their systems during procurement.[5, 15] While a 16GB or 24GB configuration is adequate for straightforward 4K timelines, complex workflows utilizing advanced effects or extensive multi-camera editing reliably require 48GB or 64GB of unified memory to prevent excessive swap file utilization on the internal solid-state drive (SSD).[3, 15, 16, 17]

AMD Ryzen Platforms and On-Device AI Workloads

For creators operating within the Windows and Linux ecosystems, AMD's Ryzen AI mobile processors—specifically the Strix Point architecture featured in systems like the Beelink SER9 and Geekom A9 Max—deliver a compelling alternative.[6, 18, 19] These systems are built on a hybrid core configuration, utilizing a mix of high-performance Zen 5 and highly efficient Zen 5c cores to optimize multi-threaded rendering tasks while minimizing idle power draw.[19, 20] On the graphics front, these processors integrate the Radeon 890M integrated GPU (iGPU), which leverages AMD's RDNA 3.5 architecture.[19, 21, 22] Featuring 16 Compute Units, the Radeon 890M represents a substantial leap in integrated graphics performance, delivering sufficient rendering throughput for moderate 1080p and 4K editing timelines, and exhibiting strong performance in application suites like Adobe Photoshop and Blender.[6, 23, 24] Furthermore, these platforms include a dedicated Ryzen AI Neural Processing Unit (NPU) capable of delivering up to 50 to 55 TOPS.[6, 22] This coprocessor allows creators to run localized Large Language Models (LLMs) and execute advanced AI-based image manipulation filters entirely on-device, bypassing the latency and subscription costs associated with cloud-based AI services.[6, 25, 26]

However, system memory constraints remain a key consideration.[19, 25] To maximize the memory bandwidth required by the high-performance Radeon 890M graphics core, several manufacturers opt for soldered LPDDR5X-7500 memory.[19, 21, 25] This architectural choice creates a hard ceiling on system expansion, meaning professionals working with massive datasets or local LLM inference models exceeding 30 billion parameters must purchase higher-tier configurations or seek out alternative models utilizing expandable SODIMM slots.[25]

Intel Core Ultra Platforms and NUC Architectures

Intel's Core Ultra (Series 2) processors, driving platforms like the Asus NUC 15 Pro+, represent the continuation of the legacy NUC ecosystem under the stewardship of Asus.[18, 27] These platforms combine Intel's hybrid architecture (comprising performance cores, efficiency cores, and low-power efficiency cores) with integrated Intel Arc graphics, such as the Arc 140T.[28, 29]

In workloads optimized for multi-threaded x86 performance, such as the Pugetbench Adobe Premiere Pro and Lightroom benchmarks, Intel-based NUCs demonstrate highly competitive processing speeds.[29, 30] The core architecture excels at managing complex office productivity, web rendering, and heavy database applications.[27, 30]

A primary architectural challenge for Intel-based NUCs remains thermal management under sustained heavy workloads.[1, 29, 31] When executing intense computational tasks such as 3D rendering or video encoding, the high-TDP processor can generate significant thermal energy, forcing the small cooling fan to spin at maximum speed.[29, 30, 31] This can lead to noticeable acoustic noise in quiet studio environments and, if core temperatures exceed 90

°

C, the system will dynamically throttle the CPU clock frequencies to prevent hardware damage, temporarily reducing rendering throughput.[29, 31]

Standard dual-channel DDR5 SODIMM slots allow creators to upgrade system memory up to 96GB, while also supporting high-speed PCIe Gen 5 SSDs.[27, 29]

----- **High-Bandwidth Interfaces and External Graphics Expansion**

Because the physical chassis of a mini PC cannot house a full-sized desktop discrete GPU, creators requiring elite rendering performance rely heavily on external expansion.[18, 26] The efficiency of these external setups is governed strictly by the bandwidth and latency characteristics of the interface connection.[8, 10]

The bandwidth limitations of these physical interfaces can be modeled mathematically. The total unidirectional bandwidth (BW) of a PCIe connection is determined by the number of active lanes (n) and the raw transfer rate per lane (R), adjusted for protocol encoding efficiency (e):

$$BW = n \times R \times e$$

For PCIe Gen 5, the physical transmission rate (R) is 32 GT/s per lane, and it utilizes a 128b/130b encoding scheme.[10] Thus, the efficiency factor is:

$$e =$$

$$\frac{128}{130}$$

$$\approx$$

$$0.9846$$

Applying this to a PCIe Gen 5 x8 interface (such as the MCIO 8i port) [10]:

$$BW = 8 \times 32 \text{ GT/s} \times 0.9846 \approx 252 \text{ Gbps (Unidirectional)}$$

BW

bidirectional

$$= 2 \times BW \approx 504 \text{ Gbps (Bidirectional)}$$

This corresponds directly to the advertised 512 Gbps of bidirectional bandwidth.[10]

Interface Type	Physical Connection / Protocol	Unidirectional Bandwidth (Gbps)	Real-World Performance Implications for Creators	Primary Hardware Implementations
Thunderbolt 3 / 4	PCIe Gen 3×4 Tunneling	40 Gbps (Max 32 Gbps PCIe data)	Heavy protocol encapsulation overhead; introduces notable rendering latency and up to 20% performance loss on high-end GPUs. [8, 32]	Broadly integrated across Intel NUCs, Mac minis, and high-end AMD systems. [7, 9, 11]
OCuLink (PCIe 4.0 x4)	Direct PCIe Bus Link	64 Gbps	Bypasses protocol overhead; offers lower latency and approximately double the usable bandwidth of Thunderbolt 3/4. [8]	Minisforum MS-01 (via expansion card), DEG2 Dock, AOOSTAR systems. [8, 33]

PCIe 4.0 / 5.0 x8 (EX Dock)	Direct Motherboard Riser Slot	128–256 Gbps	Provides a lossless desktop GPU connection; supports massive data transfer speeds with near-zero performance degradation. [34, 35]	Beelink GTi14 Ultra, GTi20 Ultra, paired with EX Pro Docking Station. [34, 36]
MCIO 8i (PCIe 5.0 x8)	Next-Gen Enterprise Direct Link	512 Gbps (Bidirectional)	Delivers near-native desktop performance, restricting high-end GPU performance loss (e.g., RTX 4090) to less than 2%. [10]	GPD Box Mini PC, GPD G2 eGPU Dock. [10]

Practical eGPU Implementation Topologies

To leverage external graphics processing, creators can implement three primary hardware topologies depending on their platform choice:

- Direct Bottom-Mounted PCIe Docks:** Devices such as the Beelink GTi14 Ultra feature a proprietary, physical PCIe x8 connector hidden behind a small door on the underside of the chassis.[9] When aligned and slid onto the Beelink EX Pro Docking Station, the mini PC establishes a direct, lossless hardware connection to a desktop graphics card.[9, 34, 37] Because the EX Pro Dock incorporates a built-in 600W power supply unit (PSU) and dual PCIe 5.0 signal enhancement chips, it powers and runs demanding cards like the NVIDIA RTX 4090 or RTX 5080 without requiring messy cabling or external ATX power bricks.[9, 34]
- OCuLink Open-Air Test Benches:** For workstations supporting OCuLink (either natively or via an internal M.2-to-OCuLink adapter or PCIe slot riser), open-air docks like the Minisforum DEG2 or AOOSTAR EG02 are commonly used.[8, 33, 38] In these setups, the desktop GPU is seated directly into the dock's PCIe slot and powered by a

separate, dedicated ATX or SFX PSU.[8, 33] An OCuLink cable connects the dock directly to the mini PC's motherboard.[8, 39] This open-air configuration facilitates optimal cooling and allows the creator to completely turn off the eGPU and its dedicated power supply when executing basic tasks, saving energy and eliminating a major studio heat source.[8]

- **Thunderbolt and USB4 DIY Enclosures:** Creators seeking plug-and-play convenience without modifying their system hardware utilize Thunderbolt 4 or USB4 eGPU boards, such as the EXP GDC TH3P4GAN dock.[40, 41] These docks require only a single Thunderbolt cable connection to the host mini PC and utilize the Microsoft Thunderbolt Control Center for automatic driver configuration.[41] While convenient and capable of delivering 85W of pass-through power to charge connected devices, this topology is heavily bottlenecked by protocol encapsulation overhead.[32, 40, 41] To minimize this performance penalty, it is critical that the creator connects the external display directly to the video outputs of the eGPU rather than routing the signal back to a monitor connected to the mini PC.[41]

Storage Architecture and High-Speed Network Integration

The throughput requirements of modern content creation demand a highly resilient, dual-layered storage strategy combining ultra-fast local solid-state drives with high-speed network-attached storage (NAS).[3, 16, 42, 43]

Local Storage Architectures and Redundancy

For active project files, raw camera cache, and scratch disk operations, local SSD speeds are paramount.[1, 3] Professional video editing software, such as DaVinci Resolve Studio and Adobe Premiere Pro, requires rapid read and write cycles to scrub through timelines fluidly.[3, 6] Mini PCs designed for professional use typically feature multiple internal M.2 2280 NVMe slots running over PCIe Gen 4 or Gen 5 buses, delivering sequential read speeds exceeding 7,000 MB/s.[6, 10]

To maximize local data safety and performance, advanced systems like the Minisforum MS-01 support hardware-level or software-level RAID (Redundant Array of Independent Disks) configurations.[42, 44] Creatives can configure dual drives in RAID 0 to strip data across both disks, effectively doubling read/write performance for high-bitrate raw formats, or in RAID 1, which mirrors data across both drives to protect against sudden hardware failure.[3, 42]

Furthermore, enterprise-class mini workstations can accommodate specialized U.2 NVMe SSDs via internal adapters, allowing creators to install up to 24TB of high-end, highly durable storage directly inside the system.[42, 44]

High-Speed Network Integration (10GbE NAS)

Once local projects are completed or require archiving, they must be moved to central storage.[43, 45] In a collaborative studio environment, editing directly off a network-attached

storage device is the ideal workflow.[42, 46] To facilitate this without introducing latency, the mini PC must be integrated into a high-speed network.[42, 47]

While standard consumer computers are equipped with 1Gbps or 2.5Gbps RJ45 Ethernet ports, professional workstations like the Minisforum MS-01 feature dual 10Gbps SFP+ LAN ports alongside dual 2.5Gbps RJ45 ports.[42, 48] Utilizing the Intel X710 controller chip, these 10GbE SFP+ ports allow the mini PC to achieve massive network throughput.[48] This bandwidth is critical for editing 4K and 8K raw video files directly from a central NAS, bypassing local storage limitations entirely.[42, 44]

To build a high-performance 10GbE workspace, creators route optical fiber or Direct Attach Copper (DAC) cables from the mini PC's SFP+ ports to a managed 10GbE network switch, which connects directly to a multi-bay NAS enclosure equipped with high-speed SSD caching and dual-gigabit or multi-gigabit link aggregation.[42, 43, 46]

Dual-PC Streaming Infrastructure and Live Broadcast Protocols

For live broadcasters and gaming content creators, dual-PC setups are highly effective.[2, 49] By separating the computational workloads of gameplay rendering and video broadcasting across two distinct machines, creators can maximize graphics settings on the primary gaming PC while ensuring a perfectly stable, high-fidelity stream on the secondary broadcast machine.[2, 47]

Hardware vs. Software Video Routing

To transmit the real-time video and audio signals from the primary gaming system to the secondary SFF streaming PC, creators can implement either a hardware-based or software-based routing topology:

- **Hardware-Based Capture Card Topology:** This configuration offers maximum reliability and zero computational impact on the gaming PC.[2, 49] The creator connects the HDMI/DisplayPort output of the gaming PC's GPU directly to the input of a dedicated capture card (such as the Elgato HD60 X).[2, 50] The capture card, which can be an external USB device connected to the streaming mini PC or an internal PCIe card housed within a mini workstation like the Minisforum MS-01, captures the video frame-by-frame and presents it as a direct camera source in Open Broadcaster Software (OBS Studio).[2, 42, 51]
- **Software-Based NDI Protocol Topology:** Network Device Interface (NDI) is a royalty-free software protocol developed by NewTek that transmits high-definition, low-latency video and multi-channel audio over a local area network.[47, 49] By installing NDI tools and the DistroAV plugin in OBS Studio on both the gaming and streaming machines, the primary system can broadcast its main output stream across the local network.[47, 50, 51] The secondary streaming mini PC then captures this network feed natively.[47, 50] While this setup completely eliminates the need for expensive capture cards and complex display cabling, it requires high network stability.[47, 49] Both computers must be hardwired via Cat6 or Cat7 cables to a dedicated gigabit network

switch, and the gaming PC will experience minor CPU and network overhead to compress and packetize the NDI stream.[47, 49]

Encoding Mechanics on SFF Systems

Once the video signal is received by the streaming mini PC, the system must encode the raw feed into a highly compressed format (typically H.264, HEVC, or AV1) for transmission to streaming platforms.[2, 11] SFF systems excel in this role by leveraging built-in hardware encoders:

- **Intel QuickSync (QSV):** Integrated into Intel Iris Xe and Arc graphics architectures, QuickSync handles high-bitrate video encoding with near-zero CPU utilization, delivering excellent stream quality on budget-friendly Intel mini PCs.[2, 42, 44]
- **NVIDIA NVENC:** If the streaming mini PC is connected to an external NVIDIA GeForce GPU via OCuLink or a physical docking station, it gains access to the NVENC encoder.[2, 8, 34] Renowned for its superior encoding efficiency, NVENC matches the visual quality of CPU-heavy x264 Slow presets without introducing processing bottlenecks.[49, 50]
- **AMD AMF:** While AMD’s Advanced Media Framework handles basic encoding tasks, independent evaluations note that it can exhibit lower encoding efficiency and higher performance overhead compared to Intel QuickSync or NVIDIA NVENC, making Intel or NVIDIA-assisted configurations the preferred choice for professional live broadcasters.[49]

Comprehensive Specifications and Performance Evaluation of Creative SFF Systems

To assist creative professionals in selecting the optimal system for their specific workflows, the following matrix compares the technical specifications, physical ports, thermal characteristics, and baseline pricing of leading SFF and NUC architectures:

Platform Model	CPU Architecture	iGPU/GPU Configuration	Expandable RAM Limit	High-Speed Storage Capability	Core I/O Ports	Thermal/Acoustic Profile	Baseline Price
Apple Mac mini M4 Pro	Apple M4 Pro (12-Core CPU, 16-Core NPU) [11]	16-Core GPU (Hardware Ray Tracing) [5, 11]	Up to 64GB Unified Memory (Soldered) [11]	Up to 8TB SSD (Soldered, ~6,000 MB/s) [3, 11]	3× Thunderbolt 5, 2× USB-C, 1× HDMI [11]	Exceptionally quiet (5 dBA at idle, no high-pitch fan whine) [11, 52]	\$1,399 [3]

Asus ROG NUC (2025)	Intel Core Ultra 9 275HX [53]	NVIDIA GeForce RTX 5080 Laptop GPU (16GB GDDR7) [53]	Up to 96GB DDR5-5600 [53]	2× M.2 NVMe (1x PCIe Gen 5, 1x PCIe Gen 4) [53]	1× Thunderbolt 4, 4× USB-A, 2× DP 2.1, 2× HDMI 2.1 FRL [53]	Unified Vapor Chamber; fans become audible under heavy load [18, 53]	\$1,629 [54]
Beelink SER9 Pro	AMD Ryzen AI 9 HX 370 (12-Core, 24-Threads) [6, 21]	AMD Radeon 890M (16 CUs, 2900 MHz) [6, 21]	32GB LPDDR5 X-7500 (Soldered) [6, 21]	2× M.2 2280 PCIe 4.0 x4 (Up to 8TB total) [6, 21]	1× USB4, 1× DP 1.4, 1× HDMI 2.1, 1× 2.5GbE [21, 55]	MSC 2.0 Vapor Chamber; quiet 32dB operation at 65W TDP [6, 21]	\$999 [36]
Asus NUC 15 Pro+	Intel Core Ultra 9 285H [28, 30]	Intel Arc 140T [28, 30]	Up to 96GB DDR5-6400 [27]	1× M.2 2280 PCIe Gen 5 x4 SSD [27, 29]	2× Thunderbolt 4, 3× USB-A, 2× HDMI 2.1, 1× 2.5GbE [27, 29]	Runs hot (up to 96°C under load); fan is noticeably loud [29, 31]	\$1,199 [7]

Geekom A9 Max	AMD Ryzen AI 9 HX 370 [18] or HX 470 [22]	AMD Radeon 890M [18, 22]	Up to 128GB DDR5 [26]	1× M.2 2280 (up to 4TB), 1× M.2 2230 [12]	2× USB4, 5× USB-A, 2× 2.5GbE, 2× HDMI 2.1, SD Card Reader [7, 22, 26]	Efficient cooling but vents emit warm air; single-channel RAM bottlenecks stock performance [22, 26]	\$1,699 [18]
Minisforum MS-01	Intel Core i9-13900H (14-Core, 20-Threads) [42, 48]	Intel Iris Xe Graphics [42, 48]	Up to 64GB DDR5-5200 [42]	3× M.2 NVMe (or U.2 Enterprise SSD support) [42]	1× PCIe 4.0 x16 (runs at x8), 2× 10G SFP+, 2× 2.5GbE, 2× USB4 [42, 48]	Smart fan curves; dual SFP+ Intel X710 chips feature independent heatsinks [46, 48]	\$775 [44]

Budget and Entry-Level Alternative Platforms

For creators with modest workload requirements or strict budget limitations, entry-level SFF platforms present highly cost-effective options.[12, 56, 57] While these models lack dedicated discrete graphics or next-generation expansion buses, they offer robust performance for daily content creation tasks, light 1080p video editing, and digital illustration.[12, 56, 57]

- **Geekom A5 (2025 Edition):** At approximately \$300, this platform is powered by an AMD Ryzen 5 7430U CPU, features dual-channel memory slots, and supports up to four external monitors.[12] This makes it an ideal choice for multi-monitor office management or budget-conscious streaming setups.[12]
- **GMKtec NucBox M6 Ultra:** Priced between \$390 and \$470, this machine is equipped with an AMD Ryzen 5 7640HS or Ryzen 7 7640HS CPU, includes 32GB of DDR5 RAM, a 1TB SSD, and dual 2.5GbE network connections out of the box.[56, 57] It delivers strong multi-core processing speeds for casual creators.[56, 57]

- **AceMagic K1:** Utilizing either an Intel Core i5-12600H or AMD Ryzen 5 7430U, this system supports triple 4K monitors and is expandable up to 64GB of RAM and 4TB of local storage.[56] This provides an expandable foundation for home studios.[56]
- **Kamrui Hyper H2:** Featuring an Intel Core i5-14450HX high-performance mobile processor, 32GB of RAM, and a 1TB SSD for \$568, this NUC alternative delivers substantial multi-core processing capabilities for intensive rendering tasks.[57]
- **GMKtec K15 AI:** Populate with 48GB of DDR5 memory and an Intel Core Ultra 5 125U processor for \$870, this system integrates an OCuLink expansion port directly into its chassis.[57] This enables creators to construct a high-performance external GPU workstation on a budget.[57]

Strategic Studio Implementation Recommendations

To construct a highly efficient, space-saving production studio, creative professionals must match their hardware topography with the technical demands of their primary software suite.

Professional Post-Production and Color Grading

For studios whose daily deliverables are processed through Apple Final Cut Pro or DaVinci Resolve Studio, a configuration based on the Apple Mac mini M4 Pro is highly effective.[3, 5, 15] The unified memory bandwidth of the M4 Pro chip easily handles 10-bit 4K footage captured in Canon Clog 3 or Sony XAVC-S without requiring proxy generation.[16, 17] To protect against timeline lag when utilizing heavy noise reduction or temporal filtering, the platform should be specified with at least 48GB of unified memory.[3, 17] Active projects should be edited on a high-speed external Thunderbolt 5 SSD array configured in a RAID 0 striped array, allowing the internal system drive to remain unburdened.[3, 16]

Real-Time 3D Rendering and Local AI Development

When creative pipelines rely on CUDA-accelerated processing in Blender, Maya, or Unreal Engine, or require running dense, localized 30-billion-parameter language models, an x86 platform with a dedicated external GPU path is mandatory.[8, 25, 34] The Beelink GTi14 Ultra paired with the EX Pro Docking Station provides a robust, direct-to-bus solution.[34, 36] By sliding the mini PC onto the docking station's physical PCIe x8 connector, the system accesses desktop discrete GPUs (such as the NVIDIA RTX 5080 or RTX 5090) with virtually zero performance loss, bypassing the major bandwidth and latency bottlenecks typical of traditional Thunderbolt enclosures.[9, 34, 35] The system should be specified with 96GB of dual-channel DDR5-5600 memory to handle massive scene geometries and large AI context windows.[9, 34, 36]

Collaborative LAN Workstations and Network Editing

In multi-editor post-production environments where assets are stored on a central server, SFF machines must prioritize network throughput over local GPU power.[42, 46] The Minisforum MS-01 is specifically engineered for this workflow.[42, 48] Powered by its dual 10Gbps SFP+ network ports driven by the dedicated Intel X710 chip, editors can connect directly to a high-speed corporate fiber switch.[42, 46, 48] This enables seamless, real-time video editing of

raw camera files directly off a central NAS array, removing the need to copy massive projects locally and simplifying the collaborative post-production workflow.[42, 44, 58]

Live Broadcasting and Broadcast Streaming

For live broadcasters executing high-bitrate streaming to Twitch, YouTube, or Kick, a dual-PC topography remains the industry standard.[2, 49] The primary gaming or rendering PC should route its video feed to a secondary, dedicated SFF streaming machine like the Beelink SER9 Pro.[2, 6, 21] By utilizing an external capture card like the Elgato HD60 X configured to duplicate the gaming display, the stream is captured by the streaming PC running OBS Studio.[2, 50, 51] The SER9 Pro utilizes its integrated AMD Ryzen AI 9 HX 370 processor and dedicated hardware encoders to compress and broadcast the feed.[6, 21, 55] This offloads the entire encoding burden from the primary gaming machine, ensuring maximum frame rate stability while maintaining an ultra-quiet 32dB acoustic profile in the studio.[2, 6, 21]

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